

DUSTON MOORE, PhD

Formal Methods | Python Tooling | Auditable AI-Assisted Engineering

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OPEN TO RELOCATION TO OTTAWA OR ANOTHER WORK LOCATION WITH EMPLOYER-SUPPORTED RELOCATION

PROFESSIONAL PROFILE

Independent researcher and software developer building exact, independently checkable tools for AI-assisted engineering and distributed systems. Experience designing Python command-line tools and packages, deterministic serialisation, exact rational computation, structured certificates, test suites, tamper rejection, provenance handling, and fail-closed verification, supported by Rocq proofs and OCaml implementations. Former tenured associate professor and Director of General Education with extensive experience translating complex standards into reviewable workflows, aligning expert groups, documenting technical decisions, and maintaining human accountability.

TECHNICAL AND SYSTEMS-ENGINEERING CAPABILITIES

Languages and tooling: Python, OCaml, Rocq/Coq, shell scripting, LaTeX, Git and GitHub, JSON.

Software engineering: CLI and library API design, packaging, automated testing, CI-friendly execution, deterministic artefacts, exact rational arithmetic, schemas, fixtures, and technical documentation.

Assurance and governance: Independent verification, trust-boundary design, provenance, admissibility gates, traceability, tamper rejection, fail-closed behaviour, and requirements-to-test correspondence.

AI-assisted engineering: Structured evaluation workflows, human-in-the-loop review, validation gates, source discipline, baseline quality, requirements clarity and verifiability, and change-impact reasoning.

Adoption and communication: Pilot demonstrations, stakeholder consultation, training, guideline development, cross-disciplinary facilitation, and concise executive and technical reporting.

SELECTED TECHNICAL WORK

Proof-Carrying Exactness | github.com/dhwcmoore/proof-carrying-exactness

Python verifier, untrusted generator, and end-to-end assessment boundary for four exact-rational verdicts.

Separates an untrusted certificate generator from a pure production verifier and independently re-verifies generated output before release.

Supports EXACT, UNDERDETERMINED, OBSTRUCTED, and INADMISSIBLE certificates, deterministic bytes, tamper rejection, and fail-closed CLI behaviour.

Maintains explicit traceability between semantic specifications, certificate obligations, implementation boundaries, tests, and non-claims.

Regional Obstruction Calculus | github.com/dhwcmoore/regional-obstruction-calculus

Exact-rational methods for detecting and auditing local-to-global coherence failure in finite distributed systems.

Combines Python diagnostics, Rocq machine-checked results, OCaml verifiers, and extracted exact-rational witness generation.

Produces constructive repair or separator witnesses rather than unaudited PASS/FAIL assertions.

Distinguishes preservation from reflection so that lossy presentation changes cannot silently conceal inconsistency.

Structural Admissibility in Rocq | github.com/dhwcmoore/structural-admissibility-in-rocq

Formalisation of observational admissibility, warrant debt, non-factorisation, and rupture decomposition.

Rocq proofs establish that distinctions destroyed by an observation map cannot be recovered by reasoning confined to that observational regime.

OCaml components realise staged observations, witness detection, and refinement loops with executable tests.

Lift-Descent Exactness | github.com/dhwcmoore/lift-descent-exactness

Constructive Rocq mathematics for lifting, claim descent, canonical values, ambiguity, and presentation transport.

Formalises exact, underdetermined, and obstructed outcomes with constructive separator and gauge witnesses.

Provides the common mathematical layer behind the regional and proof-carrying work.

PROFESSIONAL EXPERIENCE

Independent Researcher, Formal Methods and AI Systems | Edmonton, Alberta | 2023-Present

Architect and maintain a multi-repository research and software programme spanning formal specification, executable Python and OCaml tools, Rocq proof developments, independent verifiers, tests, reproducible demonstrations, and public documentation. Translate difficult questions about distributed evidence, requirements, admissibility, ambiguity, and change into explicit schemas, proof obligations, command-line workflows, and reviewable outputs. Use AI-assisted development and evaluation while keeping generated artefacts subject to independent validation, provenance checks, and human judgement. Prepare technical proposals, research papers, architecture specifications, evaluation kits, and demonstrations for academic, industrial, and government audiences.

Education Developer, Centre for Teaching and Learning, University of Alberta | Edmonton, Alberta | 2015-2016

Consulted with faculties, departments, and individual instructors on programme design, learning outcomes, assessment, peer review, educational technology, and evidence-based improvement. Developed and delivered workshops and resources on teaching portfolios, professional ethics, experiential and project-based learning, international students, and digital learning environments. Contributed to course development, script review, and assessment for two massive open online courses and collaborated with the Arts Pedagogy Research and Innovation Laboratory.

Purdue University Fort Wayne | Fort Wayne, Indiana | 2002-2013

Associate Professor of Philosophy, tenured; Assistant Professor; Visiting Assistant Professor; Director of General Education and campus assessment leadership in concurrent appointments.

Led development and continuing assessment of a university-wide general education programme involving autonomous expert groups across Arts and Sciences, Engineering, Education, Business, and other units. Translated accreditation expectations into shared learning outcomes, review cycles, evidence requirements, corrective feedback, faculty development, and formal reporting for institutional governance. Chaired and served ex officio on the Assessment Council, reviewed annual programme reports, reported to senate, and maintained SharePoint-based course approval and formal review workflows. Designed and taught courses in logic, metaphysics, epistemology, ethics, process philosophy, and critical theory; recognised through teaching awards, nominations, and competitive grants.

EDUCATION

Doctor of Philosophy in Philosophy, Katholieke Universiteit Leuven, Belgium

Master of Philosophy, Katholieke Universiteit Leuven, Belgium

Bachelor of Philosophy, Katholieke Universiteit Leuven, Belgium

Bachelor of Theology and Religious Studies, Katholieke Universiteit Leuven, Belgium

SELECTED SCHOLARLY WORK

"Marcuse and Eternal Objects," *Process Studies* 37(2), 2008.

"Propositions: Unconscious and Non-conscious Experience," in *Applied Process Thought II*, 2009.

"Plurivocal Eros: A Metaxological Reading of Plato's Symposium," in *Between System and Poetics*, 2007.